IMO Training 2018–19

Phase 1 (Level I)

Sequences and Recurrence Relations

2018/08/07-09

I. Examples for Motivation

We motivate our discussions by the following examples.

1. A piece of paper is 1 unit thick. By folding into half, the thickness becomes 2 units. Folding into half again, its thickness becomes 4 units, and so on.

   (a) What is the thickness of the paper after it is folded 10 times?

   (b) What is the thickness of the paper after it is folded 100 times?

2. There is a circle and some diameters are drawn, splitting the circle into a number of sectors.

   (a) If 10 diameters are drawn, how many sectors are obtained?

   (b) If 100 diameters are drawn, how many sectors are obtained?

3. There is a circle and we are going to draw some straight lines to split the circle into a number of regions.

   (a) If 10 straight lines are drawn, what is the largest number of regions obtained?

   (b) If 100 straight lines are drawn, what is the largest number of regions obtained?

4. A staircase consists of n steps. A boy walks from the bottom to the top, each time climbing 1 or 2 steps.

   (a) If $n = 10$, what is the number of ways in which he can climb up the stairs?

   (b) If $n = 100$, what is the number of ways in which he can climb up the stairs?

5. Canned soft drinks are arranged in layers of hexagonal arrays. The first layer has 1 can, the second layer has 7 cans, and so on.

   (a) How many cans are there in the 10th array?

   (b) How many cans are there in the 100th array?

   (c) What is the total number of cans in the first 100 arrays?

6. For positive integer n, let

$$f(n) = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n),$$

where α and β ($\alpha > \beta$) are roots of the equation

$$x^2 - x - 1 = 0.$$

(a) Find $\alpha + \beta$, $\alpha\beta$, $f(1)$ and $f(2)$.

(b) Show that $f(n+2) = f(n) + f(n+1)$ for all positive integers n.

(c) Show that $f(n)$ is an integer for all positive integers n.

II. Finding the General Term of a Sequence from the Recurrence Relation

From the examples in the previous section, we see that it is desirable to develop an efficient way to find the general term of a sequence when a recurrence relation is known. This is the objective of this section.

A. Initial Conditions

In solving recurrence relations, some initial conditions (e.g. values of the first few terms) are necessary in order to obtain a unique answer. Without the initial conditions, we will obtain a family of solutions, or the general solution.

7. In Question 1, suppose the thickness of the piece of paper is changed to 2 mm.

(a) Do we still have the same recurrence relation?

(b) Do we still have the same answers?

8. In Question 3, suppose that in addition to the circle, we count also the outermost region (which cannot be split by the straight lines).

   (a) Do we still have the same recurrence relation?

   (b) Do we still have the same answers?

B. Arithmetic and Geometric Sequences

In case a sequence is an arithmetic sequence or a geometric sequence, the recurrence relation is simple and the general term is easy to find. The same tricks apply to sequences with similar recurrence relations.

9. Suppose an arithmetic sequence has first term *a* and common difference *d*.

   (a) Describe the recurrence relation.

   (b) Find a formula for the general term.

10. Suppose a geometric sequence has first term *a* and common ratio *R*.

   (a) Describe the recurrence relation.

   (b) Find a formula for the general term.

11. Let $a_1 = 2$ and $a_{n+1} = na_n$ for positive integers *n*. Find a_n in terms of *n*.

12. Let $a_1 = 2$ and $a_{n+1} = a_n + n^2 + n + 1$ for positive integers *n*. Find a_n in terms of *n*.

C. Method of Finite Difference

In case the general term is a polynomial, the method of finite difference provides an efficient way to compute the general term.

13. In each of the following cases, the first few terms of a sequence is given. Can you find a formula for the general term?

    (a) 1, 5, 11, 19, 29, 41, 55, …

    (b) 1, 7, 21, 49, 97, 171, 277, …

    (c) 1, 3, 39, 163, 453, 1011, 1963, …

14. Find a formula for $1^4 + 2^4 + 3^4 + \cdots + n^4$ in terms of *n*.

D. Special Tricks

While the method of finite difference is easy and simple, it only works for the polynomial case. In many situations, the general term may not be polynomials. In such cases we may still be able to find the general term by applying certain tricks.

15. Let $a_1 = 5$ and $a_{n+1} = 3a_n - 4$ for $n > 1$. Find a_n in terms of *n* using the following methods:

(Method 1) By repeatedly applying the recurrence relation.

(Method 2) By setting $b_n = a_{n+1} - a_n$.

(Method 3) By setting $a_n = b_n + k$ for an appropriate constant *k*.

E. Method of Characteristic Equations

The special tricks in the previous subsection are not so easy to apply in case of higher order recurrence equations. In such cases, the method of characteristic equations may help.

16. Consider the recurrence relation

$$a_{n+2} = 4a_{n+1} - 3a_n$$

with no initial condition given.

(a) Suppose there exist two different solutions

$$a_n = f(n) \text{ and } a_n = g(n)$$

for the general term.

(i) Is $a_n = 2f(n)$ a solution for the general term?

(ii) Is $a_n = f(n) + g(n)$ a solution for the general term?

(iii) Show that $a_n = Af(n) + Bg(n)$ is a solution for the general term for any constants A and B.

(b) How many initial conditions are needed to guarantee a unique solution?

(c) Suppose $a_n = \lambda^n$ is a solution for the general term. Find all possible values of λ.

(d) Suppose the initial conditions $a_1 = 1$ and $a_2 = 2$ are given. Find a_n in terms of n.

17. Suppose $a_1 = 1$, $a_2 = 3$ and

$$a_{n+2} = 4a_{n+1} - 4a_n$$

Find a_n in terms of n.

18. Repeat the previous question with the recurrence relation replaced by

(a) $a_{n+2} = 4a_{n+1} - 4a_n + 2n$

(b) $a_{n+2} = 4a_{n+1} - 4a_n + 3^n$

(c) $a_{n+2} = 4a_{n+1} - 3a_n + 3^n$

III. More Examples

In this section we look at some problems concerning sequences and recurrence relations from various mathematical competitions.

19. (IMO HK Prelim 1990) Let a_1, a_2, …, a_n, … be a sequence of real numbers such that $a_1 = 1$, $4a_n a_{n+1} = (a_n + a_{n+1} - 1)^2$ and $a_n < a_{n+1}$ for $n \geq 1$. Find a_n.

20. Find the number of 10-digit positive integers such that
    (a) each digit is either 1 or 2; and
    (b) there exist two consecutive 1's.

21. Let n be a positive integer. Find, in terms of n, the number of ways in which a $2 \times n$ rectangle can be partitioned into 1×2 rectangles.

22. In the decimal representation of the number

$$\left(3+\sqrt{5}\right)^{2013},$$

what is the digit immediately to the right of the decimal point?

23. Let a_n be the number of positive integers having digits 1, 3, 4 only and sum of digits equal to n.

(a) Find a_{12}.

(b) Show that a_{2n} is a perfect square for all natural numbers n.

24. (IMO 1979) Let A and E be opposite vertices of a regular octagon. A frog starts jumping at vertex A. From any vertex of the octagon except E, it may jump to either of the two adjacent vertices. When it reaches vertex E, the frog stops and stays there. Let a_n be the number of distinct paths of exactly n jumps ending at E. Prove that $a_{2n-1}=0$ and

$$a_{2n}=\frac{1}{\sqrt{2}}\left[\left(2+\sqrt{2}\right)^{n-1}-\left(2-\sqrt{2}\right)^{n-1}\right] \text{ for } n = 1, 2, 3, \ldots$$

25. (CHKMO 2003) Let *ABCDEF* be a regular hexagon of side length 1, and *O* be the centre of the hexagon. In addition to the sides of the hexagon, line segments are drawn from *O* to each vertex, making a total of twelve unit line segments. Find the number of paths of length 2003 along these line segments that start at *O* and terminate at *O*.

26. (CGMO 2008) Let the sequence x_1, x_2, …, x_n, … of positive real numbers satisfy $(8x_2 - 7x_1)x_1^7 = 8$ and

$$x_{k+1}x_{k-1} - x_k^2 = \frac{x_{k-1}^8 - x_k^8}{(x_k x_{k-1})^7}$$

for $k \geq 2$. Find positive real number a such that, when $x_1 > a$, we have $x_1 > x_2 > \cdots > x_n > \cdots$, while when $0 < x_1 < a$, the sequence is not monotonic.